



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Project Overview



Dataset

3,900 rows × 18 columns



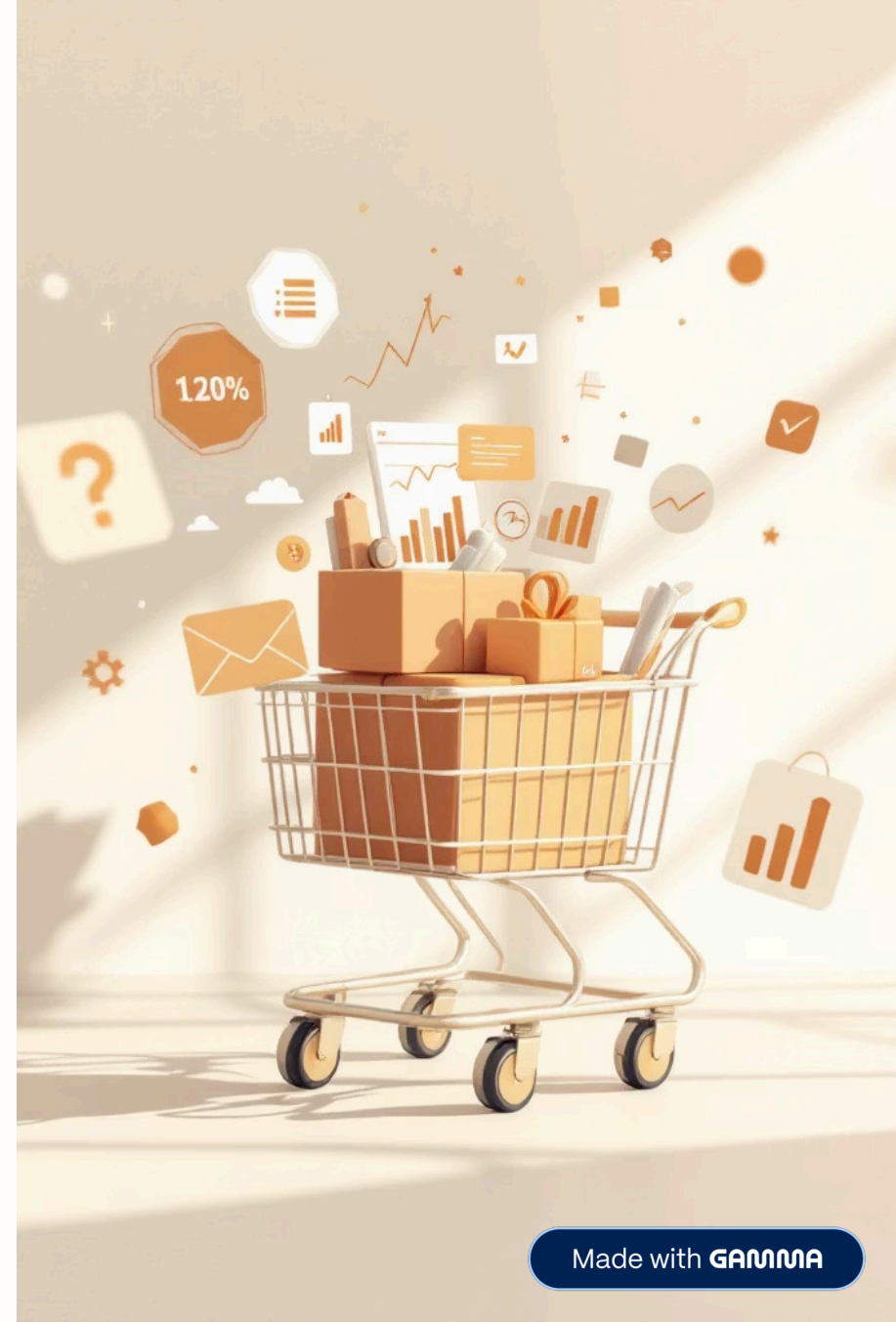
Goal

Uncover spending patterns, preferences, and behavior



Outcome

Strategic business decisions



Data Preparation & Cleaning

Python Workflow

- Data loading with pandas
- Initial exploration and summary stats
- Missing data imputation
- Column standardization

Feature Engineering

- Age group categorization
- Purchase frequency calculation
- Database integration





Revenue by Gender

\$152K

Female Customers

Higher total revenue contribution

\$148K

Male Customers

Strong purchasing power

High-Spending Discount Users

Strategy

Customers using discounts while spending above average

Insight

Discounts drive volume without sacrificing revenue



Top Products by Rating



Dresses

4.8 stars



Sneakers

4.7 stars



Handbags

4.6 stars

Shipping & Subscription Insights

Express Shipping

Higher average purchase amounts

Subscribers

Higher average spend and total revenue



Customer Segmentation



New

First-time buyers



Returning

Multiple purchases

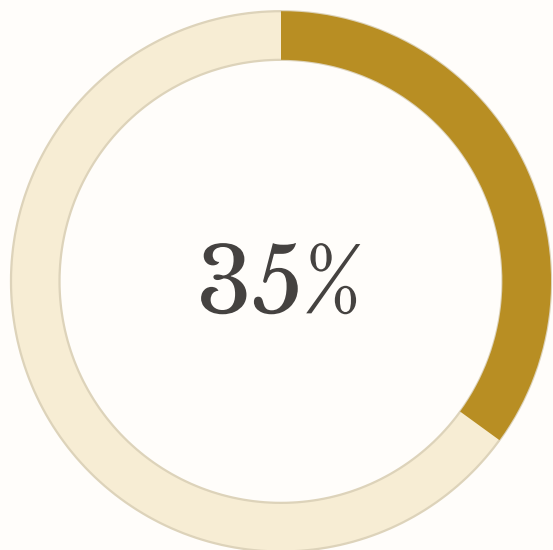


Loyal

Consistent shoppers

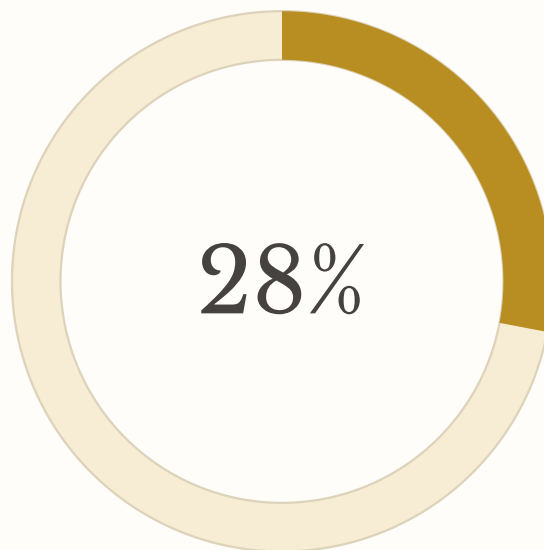


Revenue by Age Group



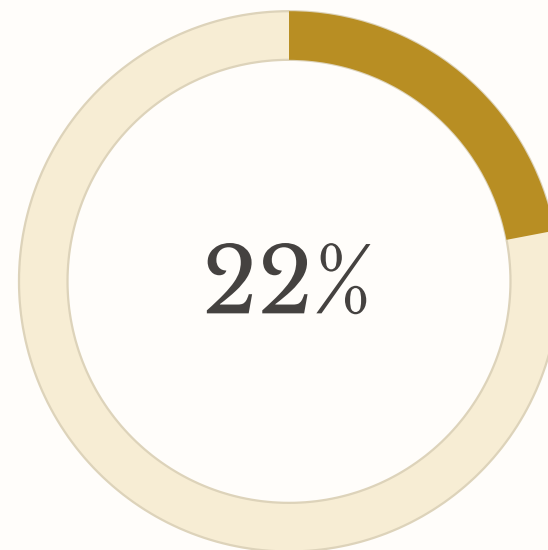
35-50 Years

Highest revenue contribution



25-35 Years

Strong purchasing power



50-65 Years

Steady spending patterns

Business Recommendations

01

Boost Subscriptions

Promote exclusive benefits

02

Customer Loyalty Programs

Reward repeat buyers

03

Review Discount Policy

Balance sales with margins

04

Product Positioning

Highlight top-rated items

05

Targeted Marketing

Focus on high-revenue segments